Arista Access Node WFD-ACC-14 Routing Protocol Flap Recovery SOP (ALARM_5093)

Node: WFD-ACC-14 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_5093

Description

This Standard Operating Procedure (SOP) outlines the steps to recover from a critical routing protocol flap failure on the Arista Access Node WFD-ACC-14 at the Whitefield POP. The failure domain involves BGP route churn, OSPF adjacency loss, and ISIS LSP flooding, resulting in the ALARM_5093 critical alarm. The root cause of this issue is likely due to a misconfiguration or hardware failure within the access node's control plane, which is responsible for maintaining routing protocol adjacencies and forwarding information base (FIB) updates. On Arista equipment, the control plane is responsible for processing routing protocol messages, installing routes into the FIB, and advertising routes to neighboring routers. A routing protocol flap occurs when the control plane experiences instability, causing the access node to repeatedly establish and teardown routing protocol adjacencies. This can lead to BGP route churn, OSPF adjacency loss, and ISIS LSP flooding, resulting in the ALARM_5093 alarm. To recover from this failure, it is essential to identify and address the root cause of the issue, which may involve inspecting the control plane configuration, checking for hardware failures, or verifying routing protocol adjacency stability.

Pre-requisite Checks

- check: Verify that the Whitefield POP is accessible and remote hands are available to perform troubleshooting and repairs., status: Complete

- check: Check the redundant link status to ensure that the access node has a stable and operational connection to the core network., status: Complete

Potential Root Causes

Cause	Probability
Misconfigured routing protocol adjacency parameters	High
Hardware failure within the access node's control plane	Medium
Routing protocol loop or oscillation	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP table version is 0, main routing table version is 0 BGP sends messages to peer-group default Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down St Pres. 192.168.1.1 4 65000 10 5 0 0 0 00:00:10 4 100 00 , Description: Verify BGP neighbor status and adjacency stability.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface 192.168.1.2 1 FULL/ - 00:00:32 192.168.1.2
```

```
Eth1/1 , Description: Verify OSPF neighbor status and adjacency
stability.

Command: show isis database detail, Expected Output: ISIS LSP Dump LSP
ID: 192.168.1.1:1 Sequence Number: 0x00000001 Checksum: 0x1234 LSP ID:
192.168.1.2:1 Sequence Number: 0x00000002 Checksum: 0x5678 ,
Description: Verify ISIS LSP stability and adjacency status.

Command: show system internal bgp summary, Expected Output: BGP
Summary Table version: 0 BGP sends messages to peer-group default
Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down St Pres.
192.168.1.1 4 65000 10 5 0 0 0 00:00:10 4 100 00 , Description: Verify
BGP summary information and routing table stability.
```

Resolution Steps

Step 1: Step: Verify and correct routing protocol adjacency parameters, Command: show ip bgp summary | include Adjacency, Description: Verify that the BGP adjacency parameters are correctly configured and that the access node is able to establish and maintain a stable BGP adjacency with the neighboring router.

Step 2: Step: Verify and correct OSPF adjacency parameters, Command: show ip ospf neighbor | include Adjacency, Description: Verify that the OSPF adjacency parameters are correctly configured and that the access node is able to establish and maintain a stable OSPF adjacency with the neighboring router.

Step 3: Step: Verify and correct ISIS LSP stability, Command: show isis database detail | include LSP, Description: Verify that the ISIS LSP is stable and that the access node is able to maintain a stable ISIS adjacency with the neighboring router.

Step 4: Step: Reset BGP session, Command: clear ip bgp *, Description: Reset the BGP session to force the access node to re-establish a new BGP adjacency with the neighboring router.

Step 5: Step: Reset OSPF adjacency, Command: clear ip ospf neighbor *, Description: Reset the OSPF adjacency to force the access node to re-establish a new OSPF adjacency with the neighboring router.

Step 6: Step: Reset ISIS LSP, Command: clear isis database *, Description: Reset the ISIS LSP to force the access node to re-establish a new ISIS adjacency with the neighboring router.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP table version is 0,
main routing table version is 0 BGP sends messages to peer-group
default Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down St Pres.
192.168.1.1 4 65000 10 5 0 0 0 00:00:10 4 100 00 , Description: Verify
BGP neighbor status and adjacency stability.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface 192.168.1.2 1 FULL/ - 00:00:32 192.168.1.2
Eth1/1 , Description: Verify OSPF neighbor status and adjacency
stability.

Command: show isis database detail, Expected Output: ISIS LSP Dump LSP
ID: 192.168.1.1:1 Sequence Number: 0x00000001 Checksum: 0x1234 LSP ID:
192.168.1.2:1 Sequence Number: 0x00000002 Checksum: 0x5678 ,
Description: Verify ISIS LSP stability and adjacency status.
```

Roll-back Procedures

- Step: Restore previous BGP configuration, Command: copy running-config startup-config, Description: Restore the previous BGP configuration to ensure that the access node maintains a stable BGP adjacency with the neighboring router.

- Step: Restore previous OSPF configuration, Command: copy running-config startup-config, Description: Restore the previous OSPF configuration to ensure that the access node maintains a stable OSPF adjacency with the neighboring router.

- Step: Restore previous ISIS configuration, Command: copy running-config startup-config, Description: Restore the previous ISIS configuration to ensure that the access node maintains a stable ISIS adjacency with the neighboring router.